

## **Version 2**

### **Naive Implementation**

#### **Objective:**

The primary objective of Version 2 is to perform a naïve GPU implementation of the convolution functions identified in Version 1. This involves:

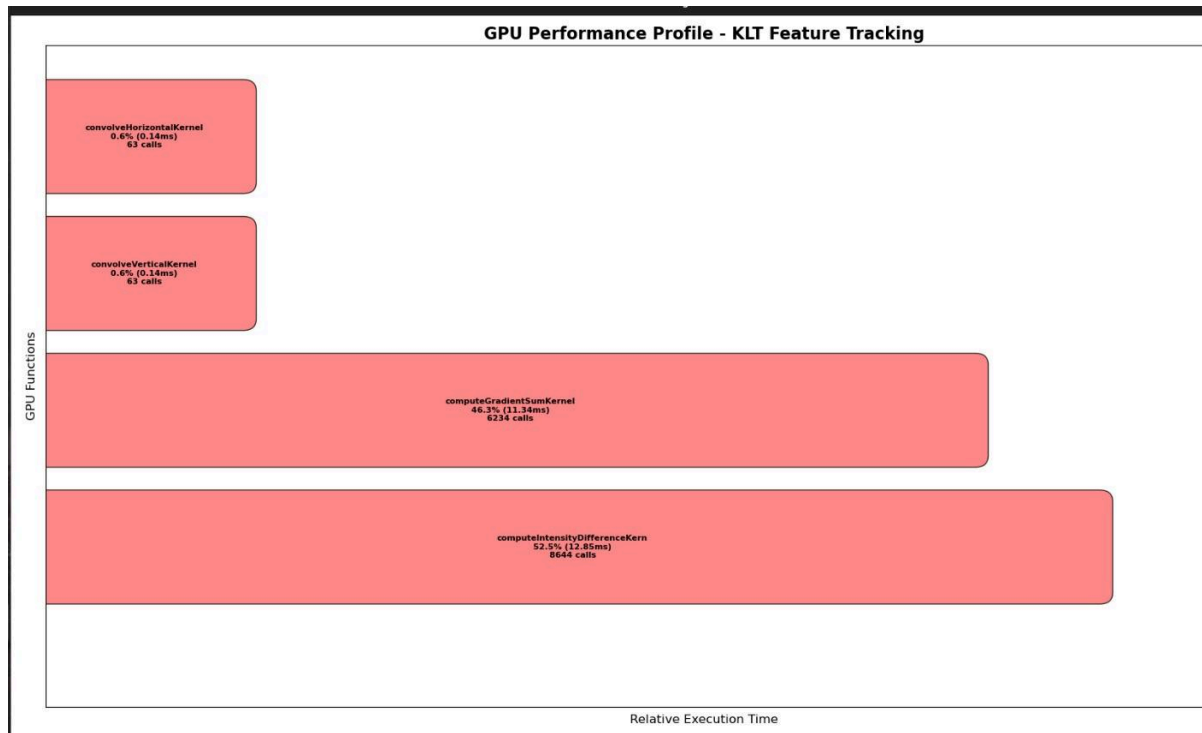
- Launching CUDA kernels to execute operations in parallel.
- Maintaining algorithmic correctness while shifting the computational workload to the GPU.

The following files are being uploaded to the GPU.

- Convole.c
- trackFeatures.c
- selectGoodFeatures.c

These contain the functions consuming most of the CPU

- \_convolveSeparate()
- \_convolveImageVert()
- \_convolveImageHoriz()
- \_computeGradientSum()
- \_trackFeatures()
- \_interpolate()



### 1. convole.cu

- convolveHorizontalKernel : (0.6% of GPU time, 63 calls)
- convolveVerticalKernel : (0.6% of GPU time, 63 calls)

### 2. trackfeatures.cu

- computeIntensityDifferenceKernel : (52.5% of GPU time, 8644 calls)
- computeGradientSumKernel : (46.3% of GPU time, 6234 calls)
- computeIntensityDifferenceLightingInsensitiveKernel
- computeWindowStatisticsKernel

### 3. selectGoodFeatures.cu

- computeMinEigenvaluesKernel
- toFloatImageKernel
- computeLocalMaxKernel

computeIntensityDifferenceLightingInsensitiveKernel, computeWindowStatisticsKernel, computeMinEigenvaluesKernel, toFloatImageKernel, computeLocalMaxKernel. These are too fast and run in microseconds, below nsys threshold that's why not visible in the graph.

**Github Link**

[https://github.com/Masooma-Ali/HPC\\_Complex\\_Computing\\_Problem](https://github.com/Masooma-Ali/HPC_Complex_Computing_Problem)